Daniel Rivera

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Education

Lakeside Technical University (Fictional) *Sep. 2018 - Jun. 2022*
BEng in Software Engineering Austin, TX
GPA: 3.64/4.00

Relevant Coursework: Databases, Operating Systems, Distributed Systems, Computer Networks

Professional Experience

Harbor Commerce (Fictional) *Jul. 2022 - Sep. 2026*
Backend Software Engineer Austin, TX

- Implemented Java services for order processing, payment-state reconciliation and inventory reservations.
- Reduced p95 API latency from 420 ms to 240 ms in staging through query plans, indexing and bounded caching.
- Added idempotency keys and transactional outbox processing; tested retries and duplicate event delivery.
- Coordinated a three-engineer migration and wrote runbooks for rollback, alerts and incident response.
- Designed explicit order-state transitions and rejected invalid updates before database writes.
- Introduced bounded connection pools and measured behavior under sustained load and downstream timeouts.
- Reviewed schema changes for compatibility with older service versions during rolling deployment.
- Investigated two staging incidents involving retry storms; added limits and dashboards for queue growth.

Clearpath Systems (Fictional) *Jun. 2021 - Aug. 2021*
Software Engineering Intern Remote

- Built REST endpoints and integration tests for a small business scheduling application.
- Added structured logging and documented error handling for client integrations.
- Worked with a product manager to translate scheduling rules into API acceptance criteria.
- Used database fixtures to test timezone boundaries, conflicting reservations and cancellation races.
- Presented a handover demo and documented setup instructions for the next intern.

Selected Projects

Reliable Task Queue *2024 - 2025*
Go, Redis, Docker

- Implemented worker leases, exponential backoff and dead-letter queues with documented delivery semantics.
- Benchmarked 1,500 jobs per second on a four-worker local setup; included hardware and payload details.
- Implemented graceful worker shutdown and tested lease expiry after forced process termination.
- Compared throughput and tail latency across different payload sizes and worker counts.
- Documented the absence of exactly-once execution and recommended idempotent job handlers.

Database Migration Guard *2023*
Python, PostgreSQL, GitHub Actions

- Built checks for migration ordering, destructive changes and backward compatibility in CI.
- Validated migration history against a clean database and a snapshot representing the previous release.
- Added a review checklist for long-running locks, backfills and rollback feasibility.
- Produced machine-readable CI summaries to help reviewers identify risky statements.

Service Load and Failure Harness *Jan. 2026 - Jun. 2026*
Go, Docker Compose, Prometheus

- Created repeatable traffic scenarios with burst loads, downstream delays and partial dependency failures.

- Recorded throughput, p95 latency, error rate and resource utilization under each scenario.
- Implemented warm-up periods and reproducible configuration files for fair before/after comparisons.
- Documented benchmark limits and avoided presenting local test throughput as production capacity.

Order State Simulation Sandbox *2025*

Java, PostgreSQL, JUnit

- Created a sandbox that replays order events under delayed, duplicated and out-of-order delivery.
- Defined invariant checks for reservation counts, cancellation states and reconciliation records.
- Compared compensating actions with database transactions in isolated failure scenarios.
- Added deterministic seeds and replayable event traces to reproduce failed tests.
- Documented which assumptions depended on external payment systems and could not be verified locally.

Incident Review Knowledge Base *2026*

Python, Markdown, Git

- Built searchable summaries of synthetic incidents for practicing operational analysis.
- Structured records around trigger, detection, customer impact, mitigation and preventive follow-up.
- Added consistency checks for timestamps and links to runbook sections.
- Led two fictional team exercises comparing escalation decisions during simulated service failures.
- Published a contributor guide emphasizing reproducible evidence and blameless incident descriptions.

Technical Skills

Programming Languages: Java, Go, Python, SQL

Frameworks & Libraries: Spring Boot, JUnit

Systems & Tools: PostgreSQL, Redis, Docker, Linux, Git

Specializations: Backend services, transactional consistency, observability

Honors & Awards

Harbor Engineering Quality Award 2025 – *Recognized incident reduction and migration documentation.*

Internal Reliability Improvement Recognition 2026 – *Recognized documented failure testing and operational handover.*